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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S4 (R,S) (FT/WP) / (S2 PT) Examination April 2025 (2019 Scheme)

Course Code: EET206

Course Name: DIGITAL ELECTRONICS

Max. Marks: 100

Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

		Marks
1	Express the decimal number -26 as an 8-bit binary number in sign magnitude form, 1's complement form and 2's complement form	3
2	Convert $(100110)_2$ to BCD, Excess -3 and gray code.	3
3	With neat logic diagram, explain how OR gate can be represented using NAND gates alone.	3
4	With a neat figure, explain how basic 2-bit parallel adder is implemented using two full-adders.	3
5	Explain with neat diagram, the working of 4:1 multiplexer.	3
6	Explain how exclusive-NOR gate can be used in basic single bit comparator.	3
7	Draw the circuit of 4-bit Ring counter along with the output table.	3
8	Determine the number of flip-flops needed to design (i) a mod-2 counter (ii) a decade counter (iii) a 3 bit register.	3
9	Explain the advantage of the R-2R ladder DAC over the weighted resistor type DAC.	3
10	Give the Verilog code for (i) OR gate (ii) Half Adder	3

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

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- 11 a) What is mean by parity? Explain how the error detection is done using even and odd parity scheme with examples. (7)
- b) Explain the concept of noise margin with reference to logic families. Is it better to have a lower value of noise margin or a higher value? Which logic family (CMOS or TTL) is preferable in high noise environment and why? (7)
- 12 a) Subtract 39 from 83 using 8 bit 1's complement and 2's complement methods. Compare both the methods. (8)
- b) Explain the following performance parameters of logic families. (6)
- (i) Fanout
 - (ii) Propagation Delay
 - (iii) Power dissipation

Module -2

- 13 a) The Boolean Expression for exclusive-OR gate is $\bar{A}B + A\bar{B}$. Using this as starting point, apply De Morgan's theorems and any other Boolean laws to develop an expression for exclusive NOR gate. (6)
- b) Reduce the expression using K map in SOP form. (8)
- $Y=f(A,B,C,D) = \sum m(6,7,8,10,11,15) + d(0,2,3,4,5,9,14)$
- 14 a) Explain with neat diagram how a full adder can be implemented using half adders. (6)
- b) Obtain the simplified POS expression using K map. (8)
- $Y=f(A,B,C,D) = \sum m(1,3,7,11,15) + d(0,2,4)$

Module -3

- 15 Explain 7 segment LED display. (14)
- With relevant logic diagram, truth table and K map simplification, explain the design of BCD to seven segment decoder. Draw the detailed logic circuit diagram of the decoder.

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- 16 a) Design a 4-bit binary to gray code converter. (10)
b) Implement the function $f = \sum m(1,3,5,6)$ using 8 to 1 mux. (4)

Module -4

- 17 a) Draw the circuit diagram and timing diagram for a 3 bit ripple up counter using negative edge triggered D Flip flop. Explain. (4)
b) Design a mod 6 synchronous up counter using JK flip flops. (10)
- 18 a) Explain the procedure to convert D Flip-Flop to SR Flip-Flop with neat circuit diagram. (6)
b) Design an asynchronous decade counter using T flipflop. (8)

Module -5

- 19 a) Distinguish between Moore and Mealy machines. (4)
b) With neat diagrams and taking an example, explain the working of Successive Approximation type ADC. (10)
- 20 a) With neat diagram, explain flash type ADC. (8)
b) Differentiate between PAL and PLA. (6)
Illustrate the implementation of SOP expressions using both.
